

Calibration Checks and a Literature Recount for a Mixed Fuzzy-Dark-Matter Search Window

Stream 3 status note

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Stream 3 — Experimentation (DarkMatterK3-Home.github.io)

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What is not claimed

This note contains no dark-matter result. Nothing in it compares a model with data: no prediction is pinned for the search discussed here, no output carries a TEST or FIT label, and the comparison phases of that search are not authorized. The calibration checks are model checks of an exactly solvable Hamiltonian, not evidence about any physical system. The $K3$ surfaces selected by the geometry stream, including X_3 and X_4 , map to no dark-matter observable, and no number from that work is used here as a prior, a parameter or an input.

Abstract

We report the state of the programme’s experimentation stream on 2026-10-08. Two numerical calibrations pass on finite Fibonacci rings of the Aubry–André chain. The exact duality between couplings λ and $4J^2/\lambda$ holds with errors below 10^{-11} at $N \leq 233$, and a seeded random potential of the same strength breaks it, as required. The extended and localized regimes and their crossing at $\lambda = 2J$ hold for $N \leq 987$ outside a band $|\lambda - 2J| < 0.15$, which is reported and not claimed. Separately, a literature recount was carried out for a proposed search for a sub-dominant fuzzy-dark-matter component with fraction $f < 1$ in the one-dimensional Lyman- α flux power spectrum. Published mixed-fraction bounds reduce the number of grid cells that would be both decisive and still open from 221 to 182 using text-stated bounds, or to 152 when figure-read contours are included. The proposal’s stop condition does not fire. No published analysis of the DESI DR1 spectrum for a mixed fraction was found.

1 Where the stream stands

The stream’s governing rule is that no real-data comparison is coded before a prediction is pinned. Two of its routes are stopped: a two-dimensional transverse search is closed by the data floors of public surveys, and a flux/tadpole route is blocked until a threefold base is specified. The open direction is an exclusion-type search over a grid of fuzzy-dark-matter masses m and fractions f . Under its proposal, Phase 0 is a literature re-survey; Phases 1–4 build and run the comparison. The decision authority approved Phase 0 only, on 2026-10-08. A noise-calibration experiment (E2) is held because no open noise dataset exists, and two laboratory-analogue experiments (E3, E4) remain appendices.

2 Calibration of the Aubry–André arm

The model is the tight-binding chain $(H\psi)_n = J(\psi_{n+1} + \psi_{n-1}) + \lambda \cos(2\pi pn/N) \psi_n$ on a ring of N sites, with (N, p) consecutive Fibonacci numbers so that $\gcd(p, N) = 1$. The discrete Fourier

transform intertwines $H(J, \lambda)$ with $H(\lambda/2, 2J)$ exactly, so the spectra at λ and $4J^2/\lambda$ agree after rescaling, and at $\lambda = 2J$ each non-degenerate eigenvector has equal inverse participation ratios in position and momentum. These are algebraic identities; the checker verifies them in floating point, so every verdict is $\text{PASS}(N)$ at a stated tolerance. Negative controls run first and must fail: a seeded random potential of matched strength breaks the duality, and a non-coprime p makes the checker refuse to run.

N	p	max duality gap	max intertwiner gap	control gap (random)	controls fail?	verdict
13	8	9.0×10^{-15}	2.9×10^{-14}	2.0×10^{-1}	yes	PASS
89	55	3.9×10^{-14}	1.0×10^{-12}	2.8×10^{-1}	yes	PASS
233	144	2.3×10^{-13}	7.8×10^{-12}	4.7×10^{-1}	yes	PASS

Table 1: E1, tolerance 10^{-9} . Gaps are maxima over $\lambda \in \{0.5, 1, 3\}J$; the control column is the smallest duality gap produced by the random potential, which must be large.

E1b extends the check to the whole coupling range and to the localization dichotomy, with acceptance criteria written in the checker before it ran. Extended regime: the product $N \cdot (\text{mean IPR})$ varies by less than a factor 1.5 across rings. Localized regime: the mean IPR is stable within 10% between the two largest rings and exceeds 0.1. The localized regime is evaluated at a generic phase, because at phase zero exactly degenerate mirror pairs make the eigenbasis, and therefore the mean IPR, ill-defined.

λ/J	regime	statistic over $N \in \{89, 233, 377, 987\}$	criterion met	status
0.5	extended	max / min of $N \cdot \text{IPR} = 1.024$	yes	claimed
1.0	extended	max / min of $N \cdot \text{IPR} = 1.038$	yes	claimed
1.5	extended	max / min of $N \cdot \text{IPR} = 1.056$	yes	claimed
1.9	extended	max / min of $N \cdot \text{IPR} = 1.318$	yes	near-critical: reported, not claimed
2.1	localized	$\text{IPR} \in [0.149, 0.191]$	no	near-critical: reported, not claimed
2.5	localized	$\text{IPR} \in [0.390, 0.409]$	yes	claimed
3.0	localized	$\text{IPR} \in [0.528, 0.542]$	yes	claimed
4.0	localized	$\text{IPR} \in [0.669, 0.680]$	yes	claimed

Table 2: E1b scaling ($J = 1$). The crossing of mean IPR in position and momentum changes sign at $\lambda = 2J$ for every coupling tested. Rows in the near-critical band are reported, not claimed.

These checks calibrate numerical machinery. They are not a measurement, they say nothing about any $K3$ surface, and no result here is evidence for a dark-matter model.

3 Phase 0: a literature recount for mixed fuzzy dark matter

The proposed search covers 13 masses from 10^{-22} to 10^{-19} eV and 20 fractions f from 0.05 to 1. A cell is *decisive* if an earlier pre-flight, using an optimistic proxy, finds that the data could reach two standard deviations there. A cell is *open* if no published constraint already excludes it. For $f = 1$ the window is already covered by published bounds; the question for Phase 0 was how much of the $f < 1$ region published mixed-fraction analyses have closed since the pre-flight table was written.

We re-read the sources and tabulated each bound with its locator and quoted sentence. Text-stated bounds are kept apart from contours read off figures. The latter come from a colour-thresholding script run on hash-pinned PDFs, with a reading error of about 0.03 in f . Every figure value is shifted toward closing fewer cells, and the script refuses to run unless its

reading of one contour reproduces the two limits that paper states in its text. The main sources are Kobayashi et al. (2017), for which a fuzzy component above 30% requires $m \gtrsim 10^{-21}$ eV, and Liu, Gong and Zhou (2026), whose 95% contour gives no effective limit above 10^{-20} eV.

Count	cells	P0 fires?
decisive cells (optimistic proxy)	258	—
decisive and open, before Phase 0	221	—
decisive and open, text-grade bounds	182	no
decisive and open, text and figure reads	152	no
text-grade, frequentist limits instead of Bayesian	188	—

Table 3: Phase 0 recount. “Decisive” is the pre-flight’s optimistic proxy, not a forecast. The stop condition fires if fewer than 10 cells survive or if all survivors have $f \geq 0.5$.

Headline result

Phase 0 leaves 182 decisive and open cells using text-stated bounds, or 152 with figure reads. Survivors reach down to $f = 0.05$ at most masses, so the stop condition does not fire. The DESI DR1 cosmological analysis of the one-dimensional flux power spectrum states that constraining the nature of dark matter is beyond its scope, so the proposal’s secondary trigger does not fire either.

What these counts are worth: four parallel survey reports covering CMB and large-scale structure, dwarf galaxies, reviews and the Lyman- α forest were not all audited against their sources, and nothing from them is applied. An unaudited bound, if real, could only close more cells, so the counts are upper bounds on how open the grid is. An Eridanus II star-cluster constraint is recorded and not applied, because its figure carries resonance bands and an approximation caveat near a few 10^{-21} eV.

4 What happens next, and what does not

Phases 1–4 have not started and are not authorized: they need their predecessors’ artifacts, a pinned prediction amendment, and the decision authority’s answers to five open questions of the proposal. Any output of those phases would be labeled as an exclusion or FIT, never as a TEST, until the pin. E2 stays held. The geometry stream’s results of the same date (`papers/stream2_k3_identification_twisted_route_2026_10_08.pdf` in the geometry repository) have their certificates mirrored under `data/mirrors/stream2/` for the record; they are not inputs to anything above.

Corrections recorded in band

1. Elb, first run: the localized-regime criterion failed at phase zero. The cause was eigenbasis dependence inside exactly degenerate mirror pairs, not a physics failure. The criterion was then evaluated at a generic phase; the phase-zero values are kept in the certificate.
2. Elb, second run: at $\lambda = 2.1J$ the localized criterion still fails on rings up to 987 sites. The band $|\lambda - 2J| < 0.15$ is reported and not claimed; the criterion itself was not changed.
3. Phase 0 report draft: a sentence saying the two Lyman- α contours agreed below 10^{-21} eV was false (at 10^{-21} eV they read about 0.35 and 0.63). It was corrected before the report was merged.
4. Phase 0 figure script: the pinned hash of one PDF was first typed with a missing digit; the script's own hash check refused to run, which is the purpose of the check.
5. The repository's continuous-integration job fails at its dependency-installation step on every recent run, before any test executes. A fix is prepared and waits for a token permission. Local runs of the full test suite pass; the CI job is not green and is not described as green.

Reproducibility

- Tables: `scripts/render_stream3_note_tables.py` (`-check` fails on drift); controls in `checkers/tests/test_render_stream3_note_tables.py`.
- Calibrations: `checkers/check_aubry_andre_selfduality.py`, `checkers/check_aubry_andre_localization_scaling.py`.
- Phase 0: `scripts/wp_e6_v2_p0_overlay.py`, `scripts/wp_e6_v2_p0_digitize_figures.py`, report `briefs/WP_E6_V2_PHASE0_RESURVEY_REPORT_2026_10_08.md`.

Provenance

Generated-by: Claude (Opus 5.5), Stream 3 support | Verified-by: the renderer's `-check` and controls; each cited checker's negative controls | Reviewed-by: N (preprint; not reviewed).

References

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- [4] J. Chaves-Montero et al. (DESI), “Cosmological analysis of the DESI DR1 Ly α 1D power spectrum,” *arXiv:2601.21432*.

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